

ODELOLA SOLOMON OLUWATOBILOBA

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PROFESSIONAL SUMMARY

Senior Data Scientist and AI Engineer specialising in Generative AI, Fraud detection techniques, Large Language Models (LLMs), AI agents, computer vision, and production-scale intelligent systems. Experienced in leading the design and deployment of AI products that solve complex challenges across finance, healthcare, enterprise software, logistics, travel, fintech, and e-commerce. Published researcher with work accepted at internationally recognised AI conferences, passionate about translating cutting-edge research into scalable products that create measurable business value. Proficient in Python, PyTorch, TensorFlow, Hugging Face, MLOps, cloud platforms, and end-to-end AI system architecture.

WORK EXPERIENCE

Gopaddi

Mar 2026 - Present

Senior AI Engineer

- Engineered AI backend systems across two production Python/FastAPI codebases, spanning 20+ AI service modules, 600+ Python files, and 60+ API routes, enabling Gopaddi to deliver AI capabilities across fraud detection, travel, workspace, CRM, booking, and automation products through a unified backend platform.
- Built agentic travel booking and fraud detection infrastructure using FastAPI, LangGraph-style orchestration, dynamic LLM provider switching, and structured state machines, enabling users to complete multi-step bookings across hotels, restaurants, flights, events, nightlife, activities, travel agencies, and vacation packages through conversational AI and tracking even the smallest fraudulent transactions.
- Led the development of an LLM-powered Fraud Detection and Investigation Assistant featuring grounded risk analysis, transaction-specific Q&A, schema-validated risk assessments, prompt-injection safeguards, automated repair retries, and secure authentication forwarding, enabling fraud analysts to investigate suspicious transactions, assess fraud risk, and generate case reports without leaving the investigation workflow.
- Architected and built core capabilities for an AI-powered Fraud Investigation platform, including natural-language investigation planning, scoped investigation tool registries, permission-aware access controls, risk-based approval classification, and audit-ready case identifiers, enabling fraud analysts to convert plain-English investigation requests into secure, validated investigation workflows.
- Architected and built AI-powered Workspace Setup workflows that transform natural-language business descriptions into structured workspace and org-space proposals, stream analysis progress over Server-Sent Events (SSE), and finalize onboarding through backend API handoffs, reducing manual setup effort for new business workspaces.
- Built and integrated core AI capabilities for Gopaddi's Ally and Scout assistants, enabling text, voice, and attachment-based workplace support, AI-powered recruitment workflows, candidate profile extraction, fit scoring, interview guide generation, pre-screening analysis, email drafting, and production-ready API/database contracts, reducing recruiter screening time by 45%, improving candidate processing throughput by 3×, and accelerating frontend/backend integration by 35%.

Health Strategy and Delivery Foundation

Oct 2024 - Feb 2026

Senior AI/ML Engineer

- Led end-to-end development of domain-specific LLM-powered healthcare assistants serving 5,000 clinical users/patients across 20 deployments, improving access to evidence-based clinical guidance; the architecture was subsequently adapted into a travel-health advisory system used by 1,000,000 users, reducing query resolution time / support ticket volume by 30%.
- Architected multimodal AI agents that reasoned jointly over structured EHR/medical record data and unstructured clinical notes (processing document types and more than 100,000 records), generating personalised clinical reports for healthcare professionals and cutting manual chart-review time by 65%.
- Designed and deployed a Retrieval-Augmented Generation (RAG) pipeline over a 15,000-document/25GB clinical knowledge base, improving factual accuracy on clinical Q&A benchmarks from 78% to 92% and reducing hallucination rate by 48%, measured via human clinician review and automated faithfulness scoring.
- Directed integration of AI decision-support capabilities into a mobile healthcare app used by 50,000+ active users, driving a 20% increase in user engagement (weekly active sessions) within 4 months of launch.

Skyway Aviation Handling Company

Feb 2023 - Sep 2024

Machine Learning Engineer

- Spearheaded the design and deployment of LLM-powered conversational AI assistants supporting personalised travel planning and customer engagement, serving 1,000,000 users across 15 markets/platforms and handling 50,000+ conversations/queries per day.
- Engineered a transformer-based recommendation engine combining collaborative filtering and behavioural analytics - trained on 120 million user interactions/85 features, driving an 18% increase in user retention (30-day retention, measured over 6 months) and a 24% improvement in click-through rate.
- Built scalable inference infrastructure using Docker and Kubernetes, supporting 2,500 requests/second at 120ms p95 latency, cutting production latency 40% (from 200ms to 120ms) and infrastructure cost by 28%.
- Established MLOps pipelines automating training, validation, monitoring, and deployment across 35 models, compressing release cycles 30% (from 10 days to 7 days) and reducing production incidents by 45%.

Artificial Intelligence/Machine Learning Engineer

- Fine-tuned large-scale language models (LLMs) for multimodal travel content generation, including itinerary creation, image captioning, and language translation, across 12 content types, boosting engagement by 15% (content interaction rate) over 6 months.
- Developed LLM-powered chatbots and recommendation agents using RAG and agentic workflows, enabling intelligent social media planning and real-time guidance for 1,000,000 users, resolving 82% of queries without human escalation.
- Implemented MLOps frameworks supporting scalable deployment of AI travel assistants across web and mobile platforms, achieving 99.7% uptime and supporting 100,000 concurrent users/deployments across 3 platforms.
- Orchestrated a unified data ecosystem across cross-functional teams, integrating booking, event, and user preference data from 18 sources into a single pipeline, reducing data latency/sync errors by 45%
- Optimized continuous fine-tuning pipelines for LLMs using feedback loops and user interaction data, processing 250,000 feedback signals per day, maintaining model accuracy above 94% and cutting model drift/staleness by 38%.

Research Experience

- Ilerioluwakiiye Abolade, Emmanuel Idoko, Solomon Odelola, Promise Omoigui, Adetola Adebawo, Aondana Iorumbur, Udunna Anazodo, Alessandro Crimi, Raymond Confidence. Structure-Aware Denoising for Pediatric Chest X-Rays - MICCAI 2025. *Co-developed a dual-decoder deep learning framework for denoising low-dose paediatric chest X-rays, improving downstream pneumonia classification accuracy from 88.8% to 92.5% while preserving diagnostically relevant image structures.* - <https://arxiv.org/abs/2508.08518>
- Ilerioluwakiiye Abolade, Prince Mireku, Kelechi Chibundu, Peace Ododo, Emmanuel Idoko, Promise Omoigui, Solomon Odelola. - VAMAE: Vessel-Aware Masked Autoencoders for OCT Angiography - ICPR 2026. *Co-developed a vessel-aware self-supervised masked autoencoder for retinal OCT angiography, introducing anatomy-informed masking and multi-target reconstruction to improve representation learning and vessel segmentation performance, particularly in limited-label settings.* <https://arxiv.org/abs/2604.06583>

ACHIEVEMENTS

- Research accepted at MICCAI (Medical Image Computing and Computer-Assisted Intervention) — peer-reviewed international conference in medical AI, acceptance rate approximately 30%.
- Research accepted at ICPR (International Conference on Pattern Recognition) — peer-reviewed publication in a globally recognised computer vision and pattern recognition venue.
- Kaggle BIPOC Fellowship — Top Project Award, selected from 1,000+ participants for developing an AI framework for low-dose paediatric chest X-ray denoising to improve pneumonia diagnosis.
- Hamoye Data Science Fellowship — 2nd Best Project, out of 500+ participating teams.
- Top 5 Graduating Student, Department of Statistics, University of Nigeria, out of 120+ graduating students, 2024.

SKILLS

- **Core Expertise:** LLMs Generative AI AI Agents RAG NLP Computer Vision Recommendation Systems, DevOps, FastAPI, LLM Operations, Resilient Distributed Systems Design, Applied Statistics, Mlops
- **Programming Languages:** Python, SQL, JavaScript, C++, Laravel
- **Frameworks & Infrastructure:** FastAPI, PyTorch, PyTorch, Tensorflow, MLFlow, Hugging Face, Docker, Kubernetes

CERTIFICATIONS

- AI/ML Engineer Certification — AI Summer of Code (AISOC)
- Machine Learning Scientist with Python — DataCamp
- Data Scientist in Python — DataQuest

EDUCATION

University of Nigeria, Nsukka

2020 - 2024

B.Sc., Statistics

- **Achievements:** Second Class Honours (First Division Equivalent), CGPA: 4.03/5.00 — Ranked Top 5 out of a number of graduating students, Department of Statistics, 2024.